AN AMPLE FAMILY OF INVERTIBLE MODULES PULLS BACK ALONG AFFINE AND QUASI-AFFINE MAPS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 23337-23341 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 23343-23347, the complete author proof (3 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Sharpens the goes-up picture around the blow-up ample-family card L16087 and its spaces consumer L16104 (wave 12-13): affine/quasi-affine base change is the elementary case.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Morphisms, lemma morphisms-lemma-quasi-affine-O-ample; chapter lemma ample-family-ample-relative (S9 registered).

Source reference. Original lemma 0GTP: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 23303–23308. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0891: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 23347–23347. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be an affine or quasi-affine morphism of schemes. If Y has an ample family of invertible modules, so does X.*

Proof. By Morphisms, Lemma 0891 this is a special case of Lemma 0GTP. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.